Markdown In, HTML Out: A Crash Course

Foundations (Everyone), Chapter 4/10 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/markdown-html-crash-course

EN: Every format decision answers one question: who reads this in the end? A human in a browser -> HTML. An AI (including a future chat) -> Markdown. Unsure -> Markdown. 'Write Markdown precise enough for a machine; demand HTML rich enough for a human.'
Roman Urdu: Har format decision ek sawal se decide hota hai: yeh aakhir mein kaun parhega? Insaan browser mein -> HTML. AI (future chat sameet) -> Markdown. Pata nahi -> Markdown. 'Machine ke liye precise Markdown likho; insaan ke liye rich HTML maango.'

1. The Two Languages

- Markdown = the specification language -- agents are heavily Markdown-trained. '#' heading = a hierarchy signal, not decoration. Writing an explicit 'Out of scope' section catches constraints prose usually buries.
- **Asymmetry:** You -> Agent = Markdown (unambiguous, fast). Agent -> You = HTML (rich, readable, shareable). Agent -> Agent = Markdown (compact, precise).
- HTML's condition: the reader must be human. For AI, HTML is noise -- tags cost thousands of words Markdown says in 50.

2. Markdown, the Writing Language

- **Headings:** one '#' title per doc, never skip levels, headings make claims not labels ('Budget: PKR 50,000 hard ceiling' beats 'Budget').
- **Lists:** bullets = unordered set, numbers = sequence that matters. Checkable bullets beat vague adjectives.
- **Code fences:** signal data vs instruction (not a safety lock) -- fencing one example output row beats describing format in 3 paragraphs.
- **First complete spec skeleton:** Goal / Context / Requirements / Hard constraints / Out of scope / Expected output. Validate-before-build: have the agent attack its own spec, 2-3 rounds.

3. HTML, the Reading Language

- HTML adds 5 things: density, navigability, shareability, interactivity, a copy-back loop -- at the honest cost of more tokens.
- **5 HTML patterns:** comparison grids, explainers/reports, code review, live-control prototypes, disposable throwaway tools.
- **Publishing ladder (easiest to most durable):** Claude.ai publish -> GitHub Gist -> GitHub Pages -> Netlify.
- Document destination depends on what the reader will DO with it: sign->PDF, edit->Word, present->Slides, numbers->Excel, machine->CSV.

4. One Exercise, Three Motions

- Same spec.md -> report.html exercise run 3 ways: Chat (paste-based), Terminal (Claude Code/OpenCode gathers its own context), Desktop (Cowork/OpenWork plans before touching files).

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Markdown (Intent Layer)	The specification language -- unambiguous, machine-parsed.	Specification language -- unambiguous, machine-parsed.
Asymmetry rule	You->Agent and Agent->Agent = Markdown; Agent->You = HTML.	You->Agent aur Agent->Agent = Markdown; Agent->You = HTML.
Out of scope section	An explicit heading naming what's deliberately excluded.	Explicit heading jo deliberately excluded cheezein naam kare.
Checkable bullet	A requirement phrased so pass/fail is objectively testable.	Requirement jo objectively pass/fail test ho sake.
Publishing ladder	Claude.ai publish -> Gist -> GitHub Pages -> Netlify (increasing durability).	Claude.ai publish -> Gist -> GitHub Pages -> Netlify (barhti durability).

Quick Self-Check

Question	Answer
What single question decides Markdown vs HTML?	Who reads this in the end -- a human (HTML) or an AI, including a future chat (Markdown)?
Why does HTML count as 'noise' for an AI reader?	Tags and styling cost thousands of words to say what Markdown says in 50.
What's the 'validate-before-build' move for a spec?	Have the agent attack its own spec for 2-3 rounds before building, improving it from a 6 to a 9.
Why fence one example output row instead of describing the format in prose?	A concrete fenced example removes ambiguity that 3 paragraphs of description would still leave open.
How do you decide a document's final format (PDF vs Word vs CSV)?	By what the reader will DO with it: sign->PDF, edit->Word, present->Slides, machine-reads->CSV.

Where This Leads / Sources

Leads to: Code You Never Write, Skills & Connectors (SKILL.md is Markdown). Full appendix: Al-Noor Tuition Academy worked example (spec -> ~6KB HTML).

Source: docs/markdown-html-crash-course/ (this repo), content originally via Zia Tutor AI connector.